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Study of IR Photoheating of Aqueous Solutions of Boron Nanoparticles Synthesized by Pulsed Laser Ablation for Cancer Therapy

I. B. Belyaev[a,b], I. V. Zelepukin[a,b], A. I. Pastukhov[c], P. V. Shakhov[a], G. V. Tikhonowski[a], A. A. Popov[a], A. Yu. Zakharkiv[a], S. M. Klimentov[a], A. A. Garmash[a], I. N. Zavestovskaya[a,d], S. M. Deyev[a,b], and A. V. Kabashin[c,*]

[a] *National Research Nuclear University "MEPhI," Engineering Physics Institute of Biomedical Chemistry, Moscow, 115409 Russia*

[b] *Shemyakin and Ovchinnikov Institute of Bioorganic Chemistry, Russian Academy of Sciences, Moscow, 117997 Russia*

[c] *LP3, Aix Marseille University, CNRS, 13288 Marseille, France*

[d] *Lebedev Physical Institute, Russian Academy of Sciences, Moscow, 119991 Russia*

**e-mail: andrei.kabashin@univ-amu.fr*

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Abstract—Photothermal properties of boron (B) nanoparticles (NPs) synthesized by pulsed laser ablation in liquid (PLAL) are studied. It is found that B NPs have high optical extinction coefficient in the near infrared (IR) region, in particular, its value was 0.2 l/(cm g) at a wavelength of 800 nm. It is shown that aqueous suspensions of B NPs can be heated during laser irradiation with a wavelength of 830 nm and a radiation power of 1 W. Here, the temperature increase varied from 6.5 to 36°C under the increase of the particle concentration from 0.04 to 0.8 g/L; while the relevant parameter in the absence of particles was 2°C. B NPs were able to provide a stable photothermal response during several irradiation cycles. The results evidence a high potential of B NPs as sensitizers of photothermal therapy (PTT).

Keywords: laser ablation in liquid, photoheating, optical spectroscopy, nanoparticles, boron

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1. INTRODUCTION

Nanomedicine is an actively developing topic that focused on the use of a variety of nanomaterials for diagnostics and therapy of various diseases [1, 2]. Among these nanomaterials, nanoparticles (NPs) of boron (B) and its compounds look especially promising for medical applications due to their ability to provide a high concentration of active boron isotopes for neutron-capture therapy [3–5].

In addition, B NPs look promising for photohyperthermia (phototherapy), which is another appealing method for cancer treatment based on local thermolysis of tumorous tissue via the employment of nanodrugs having a high rate of conversion of absorbed light to heat [7]. To treat deep tissues, photothermal therapy (PTT) requires effective sensitizers of light absorption in the near infrared region (NIR) to match the window of relative biological transparency. However, to the best of our knowledge, the photothermal capability of B NPs has not yet been explored in detail in the literature.

This paper presents a study of photothermal properties of B NPs in the first window of relative transparency of biological tissues (650–950 nm). Boron NPs were obtained using a recently developed laser-ablation method, which provides high surface purity and biocompatibility of nanoparticles. Studying photoinduced heating of B NPs in water using laser radiation with a power of 1 W at a wavelength of 830 nm, we demonstrated the possibility of increasing the temperature by more than 30°C at a particle concentration of 0.4 g/L. At lower concentrations, temperature conditions corresponding to milder hyperthermia modalities were achieved. The presented results can significantly increase the efficiency of nanomedicine methods using B NPs.

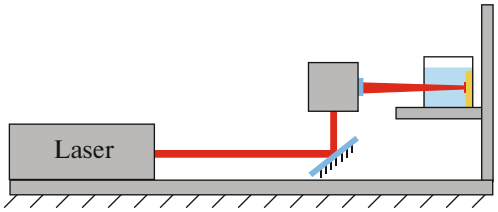

Fig. 1. Scheme of the experimental setup for PLAL.

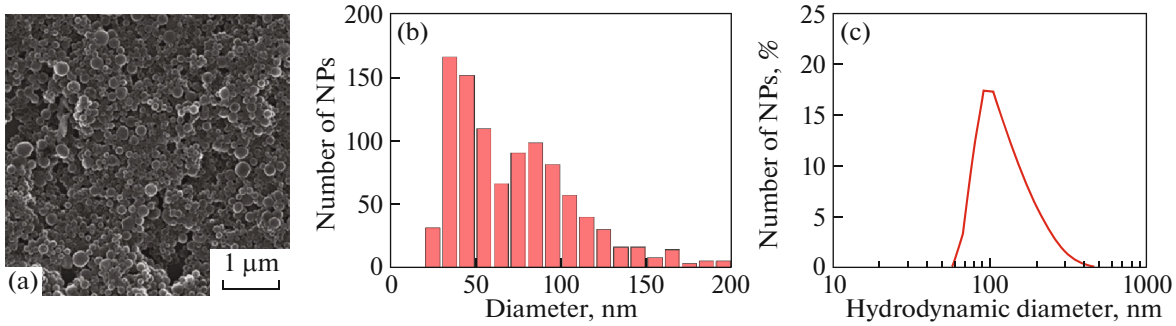

Fig. 2. (a) SEM image of B NPs. (b) Size distribution of B NPs obtained from the analysis of SEM images. (c) Hydrodynamic size of B NPs in water.

2. MATERIALS AND METHODS

Boron nanoparticles were obtained by pulsed laser ablation of a crystalline boron target (MaTecK, 99.5% purity) in deionized water (Fig. 1). One setup made possible the achievement of a high speed of the laser beam scanning relative to the target surface (up to 6000 m/s) under a high repetition rate (100 kHz) of laser pulses. During ablation, a laser beam (1030 nm, 270 fs, 50 μJ, TETA 10, Avesta, Russia) was focused on the target surface fixed vertically in a glass cuvette filled with 50 mL of liquid. The liquid layer between the cuvette wall and the target surface was 3.3 mm. The cuvette with thr target was continuously moved perpendicular to the beam.

The morphology and size distribution of B NPs were determined by scanning electron microscopy (SEM) using a MAIA 3 microscope (Tescan, Czech Republic) at an accelerating voltage of 20 kV. The size distribution was subjected to a statistical analysis by counting more than 1000 individual particles. The nanoparticle concentration in the solution was determined by the particle weight in a fixed volume after drying the liquid. Optical extinction spectra were measured using a spectrophotometer (ML 122, SOL Instruments, Belarus) in a quartz cell with an optical path length of 10 mm. The hydrodynamic size of nanoparticles was measured in water by a method of dynamic light scattering using a Zetasizer Nano ZS device (Malvern Instruments, Great Britain).

To study photothermal properties of B NPs, an aqueous solution of nanoparticles (1 mL) was placed into a plastic spectrophotometric cell with an optical path length of 10 mm, which was then irradiated with an 830 nm cw laser source through a side wall of the cell. Variations of liquid temperature were controlled using a FLIR C3 thermographic camera (FLIR, USA). Measurements were performed at an ambient temperature of 27.5°C.

3. RESULTS AND DISCUSSION

Boron NPs were obtained by femtosecond pulsed laser ablation in deionized water. Scanning electron microscopy showed that the shape of B NPs was spherical while the size distribution was bimodal with peaks around 35 and 85 nm (Fig. 2b). The hydrodynamic diameter of B NPs in water was 131±55 nm, which confirms the production of a colloidal solution of nanoscale particles.

Optical extinction spectra from solutions of B NPs had no pronounced peaks in the range of 350—1050 nm and represented a monotonically decreasing spectral dependence (Fig. 3). Nevertheless, the optical extinction coefficient was rather high in the first window of relative biological transparency (650—950 nm. In particular, this coefficient at a wavelength of 800 nm was 0.2 L/(cm g), which is comparable

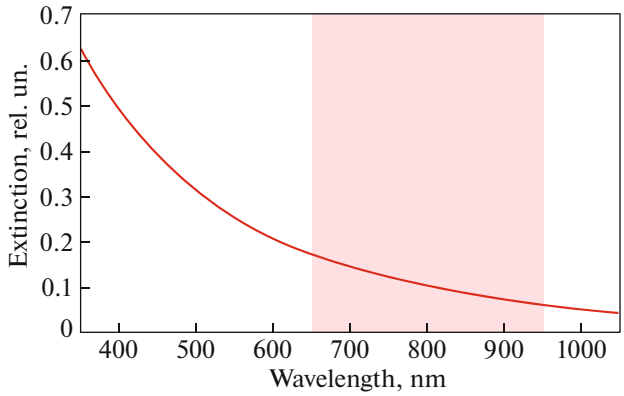

Fig. 3. Spectrum of optical extinction of B NPs. The painted area in the range of 650–950 nm corresponds to the first transparency window of biological tissues.

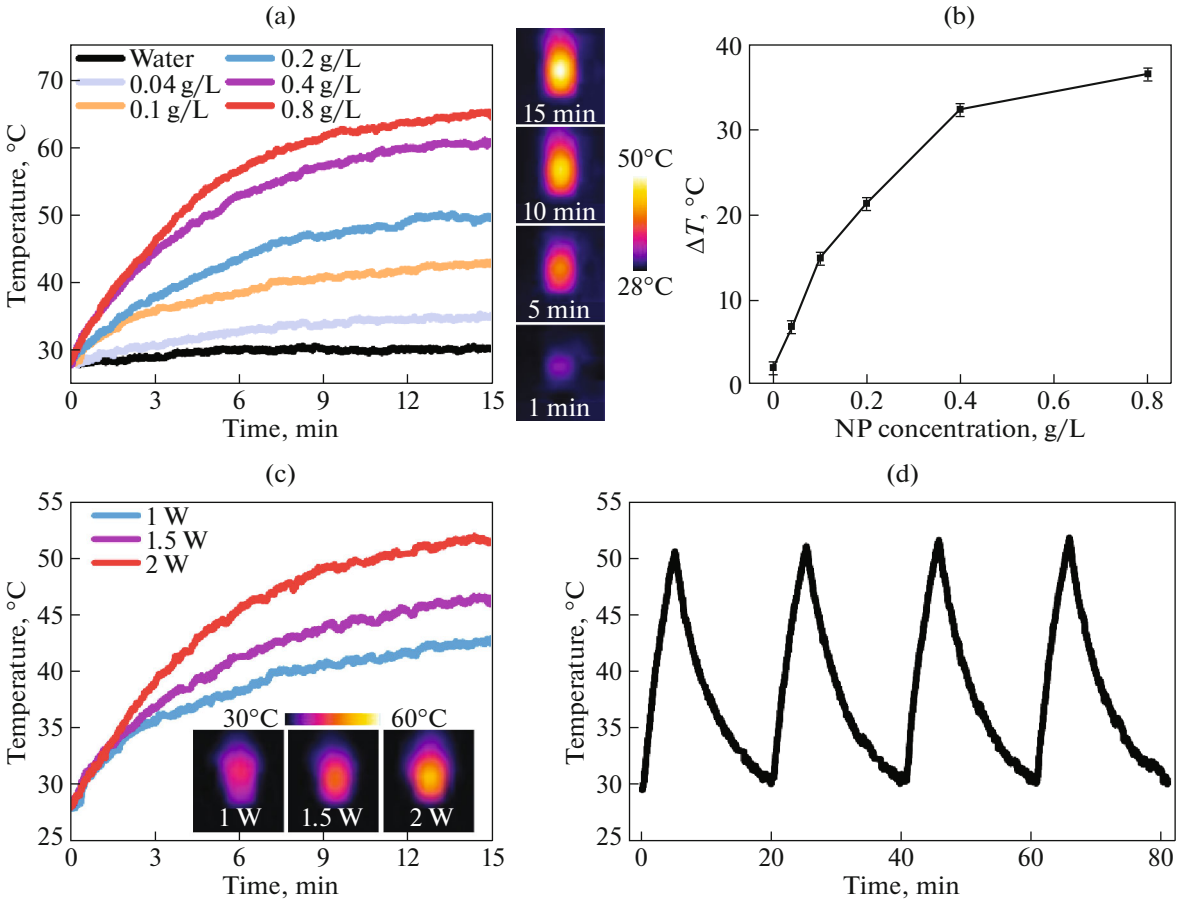

Fig. 4. Photothermal properties of B NPs during irradiation with an 830-nm laser. (a) Kinetics of photothermal heating of solutions with various NP concentrations at an incident laser beam power of 1 W. The right inset shows the variation of the temperature map of the NP suspension (0.2 g/L) during irradiation. (b) Dependence of the maximum temperature change (ΔT) for 15-min irradiation (1 W) on the NP concentration. (c) Dependences of solution temperatures on the laser radiation power at the NP concentration of 0.1 g/L. The inset shows temperature maps at maximum heating. (d) Kinetics of cyclic heating at 1-W irradiation (5 min) and cooling (15 min) of B NPs with a concentration of 0.4 g/L.

to that of nanomaterials most actively used as PTT sensitizers [7, 8]. Thus, B NPs look as promising agents to implementing photoconversion in the NIR.

Figure 4c shows a dependence of photothermal heating kinetics on laser beam power at a fixed particle concentration (0.1 g/L). One can see that the increase of radiation power from 1 to 2 W caused the increase

of steady-state temperature from 41°C to 52°C. Thus, at relatively safe irradiation intensities, B NPs can enable conditions for both "soft" hyperthermia (≥41°C) and thermal ablation of tissues (≥50°C).

It should be noted that the achievement of the therapeutic effect sometimes requires several heating cycles of the affected region; therefore, the used sensitizers should exhibit a stable photothermal effect. To study the photothermal stability, obtained NPs were subjected to cyclic photoinduced heating and natural cooling. It was found that the thermal response of B NPs conserves its properties for at least four such cycles (Fig. 4d). This high photostability distinguishes laser-synthesized B NPs from many alternative PTT sensitizers. For example, it is known that being one of the most popular PTT sensitizers, gold nanorods exhibit low photostability in the NIR [9].

Notice that the presented results continue a cycle of our studies devoted to the development of new functional nanomaterials by the PLAL method for various biomedical application [10–14].

4. CONCLUSIONS

Boron NPs were synthesized by the PLAL method in deionized water. Photothermal properties of produced NPs during irradiation with a NIR laser were studied in vitro. It was shown that temperature conditions for mild and moderate hyperthermia can be achieved at a nanoparticle concentration of ≤ 0.2 g/L, which evidences their high efficiency as photothermal agents in combined therapy. The increase of the NP concentration up to 0.4 g/L led a local temperature increase by more than 31°C, which is sufficient for thermal ablation of biological tissues. It was found that laser-synthesized B NPs demonstrate a high stability of their photothermal properties. Altogether, these data confirm the high promise of using these NPs as sensitizers for PTT applications.

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CONFLICT OF INTEREST

The authors declare that they have no conflicts of interest.

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